



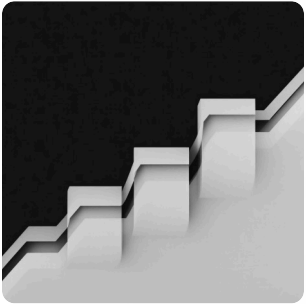
CoreModel

Your Brain Has a Fingerprint. We Map It.

An adaptive learning platform that builds a cognitive profile for every learner — then personalizes content, pacing, and coaching to match how they actually think.

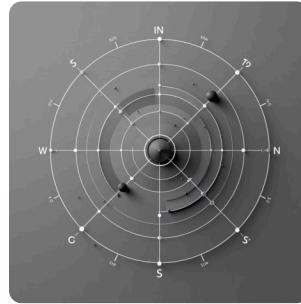


How It Works



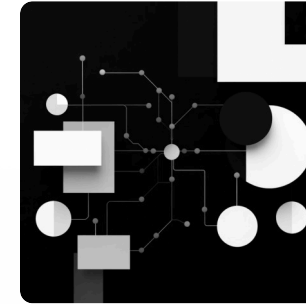
12-Screen Assessment

A quick diagnostic across 5 cognitive dimensions — takes minutes, shapes everything.



5-Dimension Analysis

Cognition, metacognition, motivation, calibration, and study habits — scored against 11 research frameworks.



Personalized Every Interaction

Chunk size, difficulty, scaffolding, and coaching tone all adapt from the first session onward.

Personalized Study Guides

Your fingerprint generates a **7-day adaptive study plan** — chunk size, difficulty, scaffolding, and coaching tone all driven by your profile.

Core Practice

60–70% of time

Metacognitive Routines

Self-awareness loops

Skill Builders

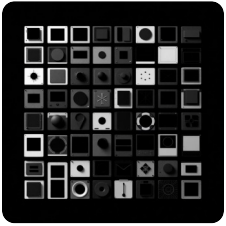
Targeted gap closure

Motivation Support

Behavioral scaffolding

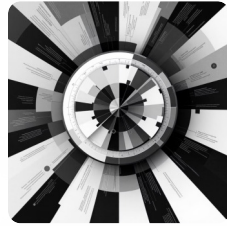


12 Artifact Types, All Fingerprint-Matched



Assessment

Quizzes & Flashcards



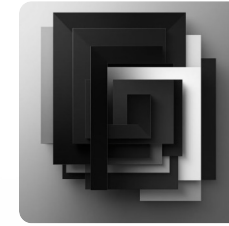
Visual

Mind Maps, Slides & Spatial
3D Visualizations



Audio & Video

Audio Lessons & Video
Explainers



Deep Practice

Worked Examples,
Interleaved Sets & Elaborative
Interrogation

Every artifact adapts **difficulty and format** to your cognitive fingerprint.

Voice + Chat: One Unified Experience

Voice

LiveKit WebRTC — real-time STT, LLM processing, TTS, 3 visualizer modes

Chat

Vercel AI SDK — streaming responses, 40+ tools for profile, guide, artifact & schedule

📄 Both channels merge into a **single conversation timeline** — nothing gets lost.

MCP Server: Your Fingerprint Travels with You

CoreModel exposes **6 learning tools** via Model Context Protocol — works inside Claude, ChatGPT, or any MCP-compatible AI assistant.



Quizzes



Flashcards



Mind Maps



Slides

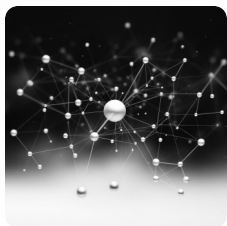


Spatial



Guides

The Stack



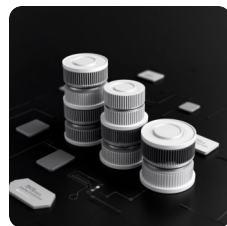
AI / ML

GPT-4o-mini · Vercel AI SDK ·
AI Gateway · LiveKit Agents



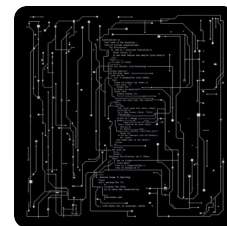
Quality

Evalite (30+ scorers) · Vitest ·
Playwright



Data

PostgreSQL (Neon) · Drizzle
ORM · Vercel KV · Vercel Blob



Frontend

Next.js 16 · React 19 · Tailwind 4
· Three.js · Deck.gl



Voice

LiveKit WebRTC · OpenAI
TTS/STT · Silero VAD



Auth & Payments

Better Auth · Stripe credits



Challenges

"The biggest challenge was translating psychological frameworks into something that actually changes AI behavior in a meaningful way — with structural differences in how information is sequenced and presented."

The DNA metaphor and gamified UI were designed to make the assessment feel **engaging, not clinical** — science that learners actually want to experience.

Team



Amy J Xu

Electrical and Computer Engineering bachelor's student at the University of Texas at Austin, passionate about AI and hardware systems. Creative and always curious to explore new technologies.



Jacqueline Azah

Master's candidate at the Texas A&M Bush School of Government and Public Service specializing in economic policy. She is interested in policy and data that shape global frameworks and improve the everyday lives of people. Her current work examines the governance of artificial intelligence and its implications for economic systems, public policy, and global development.



Gerson Mejia

Software developer with 10+ years of experience. Builder at heart. Have worked with teams of all sizes shipping products that matter. Passionate about AI and what's next.



Matt Wood

Design Engineer and product builder with 16+ years delivering digital products, including roles at JPMorgan Chase, NBCUniversal, Altana AI, and FM Global, helping teams turn complex ideas into real software used at scale.